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الجامعة الإسلامية العالمية شيتاغونغ
International Islamic University Chittagong

TITLE: STUDENT BOX

Student Management System

PROJECT REPORT

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1. Project Name

Student Box – Student Management System

2. Introduction

Student Box is a console-based Student Management System developed using the C programming language. The system is designed to simplify the management of student information in an educational institution.

The project allows users to add new student information, display all students, search students by Student ID, update student information, delete student records, calculate student grades based on CGPA, and save all data permanently using file handling.

The application uses important C programming concepts such as structures, arrays, functions, loops, conditional statements, string handling, and file handling. The project is easy to use and provides a simple menu-driven interface.

3. Motivation

Managing student records manually requires a lot of time and may lead to mistakes. The main motivation behind this project is to develop a simple digital system that can organize student information efficiently.

This project also helps students understand practical programming concepts and improves problem-solving skills using the C programming language.

4. Applications

- Add new student information.
- Display all student records.
- Search student information using Student ID.
- Update existing student information.
- Delete student records with confirmation.
- Calculate student grades from CGPA.
- Store student information permanently using text files.
- Reduce manual paperwork.
- Improve accuracy in student record management.

5. Objectives

- To develop a Student Management System using C.
- To learn structure and function implementation.
- To understand file handling techniques.
- To perform CRUD (Create, Read, Update, Delete) operations.
- To calculate student grades automatically.
- To improve programming and logical thinking skills.
- To maintain student information efficiently.

6. Source Code

The project is implemented using the C Programming Language.

Development Environment:

- Programming Language: C
- IDE: Code::Blocks
- Compiler: GCC

Functions Used:

- addStudent()
- displayStudent()
- searchStudent()
- updateStudent()

- deleteStudent()
- calculateGrade()
- saveFile()
- loadFile()

Features Implemented:

- ✓ Add Student Information
- ✓ Display Student List
- ✓ Search Student by ID
- ✓ Update Student Information
- ✓ Delete Student with Confirmation
- ✓ Calculate Student Grade
- ✓ Save Data into File
- ✓ Load Data Automatically When Program Starts

C Program Source Code:

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>

typedef struct student
{
    char id[20];
    char name[50];
    char department[30];
    char semester[10];
    double cgpa;
    char grade[5];
} STD;

void saveFile(STD student[],int count);
void loadFile(STD student[],int *count);
int checkDuplicateID(STD student[],int count,char id[]);
void displayHeader();
void showmenu();
void pauseScreen();
void addstudent(STD student[],int *count);
void displaystudent(STD student[],int count);
void searchstudent(STD student[],int count);
void updatestudent(STD student[],int count);
void deletestudent(STD student[],int *count);
void calculateGrade(STD student[],int count);
```

```
int main()
{
    STD student[100];
    int count = 0,choice;

    loadFile(student,&count);

    while(1)
    {
        displayHeader();
        showmenu();
        printf("Enter your choice: ");
        scanf("%d",&choice);
        switch(choice)
        {
            case 1:
                addstudent(student,&count);
                break;
            case 2:
                displaystudent(student,count);
                break;
            case 3:
                searchstudent(student,count);
                break;
            case 4:
                updatestudent(student,count);
```

```
        break;
    case 5:
        deletestudent(student,&count);
        break;
    case 6:
        calculateGrade(student,count);
        break;
    case 7:
    {
        char choice;

        printf("\nAre you sure you want to exit? (Y/N): ");
        scanf(" %c",&choice);

        if(choice == 'Y' || choice == 'y')
        {
            saveFile(student,count);
            printf("Data saved successfully. Thank you\n");
            exit(0);
        }
        else if(choice == 'N' || choice == 'n')
        {
            printf("Exit cancelled.\n");
        }

        break;
    }
```

```

        default:
            printf("Invalid choice! Please select 1-7.\n");
        }
        pauseScreen();
    }
    return 0;
}

void displayHeader()
{

printf("\n*****\n");
printf("          Student Box\n");
printf("International Islamic University Chittagong\n");

printf("*****\n");
}

void showmenu()
{
    printf("\n===== \n");
    printf("    Main Menu\n");
    printf("===== \n");
    printf("1. Add Student Information\n");
    printf("2. Display Student Information\n");
}

```

```

printf("3. Search Student Information\n");
printf("4. Update Student Information\n");
printf("5. Delete Student Information\n");
printf("6. Calculate Student Grade\n");
printf("7. Save & Exit\n");
printf("=====\\n");
}

void addstudent(STD student[],int *count)
{
    if(*count >= 100)
    {
        printf("Student limit reached! Cannot add more.\\n");
        return;
    }

    printf("\\n--- Add Student Information ---\\n");

    int duplicate;

    do
    {
        printf("Enter ID: ");
        scanf("%s",student[*count].id);

        duplicate =
checkDuplicateID(student,*count,student[*count].id);

```

```
    if(duplicate == 1)
    {
        printf("ID already exists! Please enter another ID.\n");
    }
}
while(duplicate == 1);

printf("Enter Name: ");
scanf(" %[^\\n]",student[*count].name);

printf("Enter Department: ");
scanf(" %[^\\n]",student[*count].department);

int sem;

do
{
    printf("Enter Semester (1-12): ");
    scanf("%s",student[*count].semester);

    sem = atoi(student[*count].semester);

    if(sem < 1 || sem > 12)
    {
        printf("Invalid Semester! Please enter a value
between 1 and 12.\n");
    }
}
```

```

    }
}
while (sem < 1 || sem > 12);

do
{
    printf("Enter CGPA (0.0 - 4.0): ");
    scanf("%lf",&student[*count].cgpa);

    if(student[*count].cgpa < 0 || student[*count].cgpa > 4)
    {
        printf("Invalid CGPA! Please enter a value between 0.0
and 4.0.\n");
    }
}
while (student[*count].cgpa < 0 || student[*count].cgpa >
4);

strcpy(student[*count].grade,"N/A");

(*count)++;

printf("Student information added successfully!\n");
}

void displaystudent(STD student[],int count)
{

```

```

    if(count == 0)
    {
        printf("No student information available.\n");
        return;
    }

    printf("\n%-12s %-30s %-15s %-10s %-5s\n","ID","Name","Department","Semester","CGPA");
    printf("-----\n");

    for(int i = 0; i < count; i++)
    {
        printf("%-12s %-30s %-15s %-10s %-5.2lf\n",student[i].id,student[i].name,student[i].department,student[i].semester,student[i].cgpa);
    }
}

void searchstudent(STD student[],int count)
{
    if(count == 0)
    {
        printf("No student information available.\n");
        return;
    }
}

```

```
char searchid[20];
printf("Enter Student ID to search: ");
scanf("%s",searchid);

int found = 0;

for(int i = 0; i < count; i++)
{
    if(strcmp(student[i].id,searchid) == 0)
    {
        printf("\n--- Student Information Found ---\n");
        printf("ID:      %s\n",student[i].id);
        printf("Name:     %s\n",student[i].name);
        printf("Department: %s\n",student[i].department);
        printf("Semester:  %s\n",student[i].semester);
        printf("CGPA:     %.2lf\n",student[i].cgpa);
        found = 1;
        break;
    }
}

if(found == 0)
{
    printf("Student with ID %s not found.\n",searchid);
}
}
```

```
void updatestudent(STD student[],int count)
{
    if(count == 0)
    {
        printf("No student information available.\n");
        return;
    }

    char updateid[20];
    printf("Enter Student ID to update: ");
    scanf("%s",updateid);

    int found = 0;

    for(int i = 0; i < count; i++)
    {
        if(strcmp(student[i].id,updateid) == 0)
        {
            printf("\n--- Update Student Information ---\n");

            printf("New Name: ");
            scanf(" %[^\\n]",student[i].name);

            printf("New Department: ");
            scanf(" %[^\\n]",student[i].department);

            int sem;
```

```
do
{
    printf("New Semester (1-12): ");
    scanf("%s",student[i].semester);

    sem = atoi(student[i].semester);

    if(sem < 1 || sem > 12)
    {
        printf("Invalid Semester! Please enter a value
between 1 and 12.\n");
    }

}
while(sem < 1 || sem > 12);

do
{
    printf("New CGPA (0.0 - 4): ");
    scanf("%lf",&student[i].cgpa);

    if(student[i].cgpa < 0 || student[i].cgpa > 4)
    {
        printf("Invalid CGPA! Please enter a value
between 0.0 and 4.\n");
    }
}
```

```
        while(student[i].cgpa < 0 || student[i].cgpa > 4);

        found = 1;
        printf("Student information updated
succcessfully!\n");
        break;
    }
}

if(found == 0)
{
    printf("Student with ID %s not found.\n",updateid);
}
}

void deletestudent(STD student[],int *count)
{
    if(*count == 0)
    {
        printf("No student information available.\n");
        return;
    }

    char deleteid[20];
    printf("Enter Student ID to delete: ");
    scanf("%s",deleteid);
```

```
int found = 0;

for(int i = 0; i < *count; i++)
{
    if(strcmp(student[i].id,deleteid) == 0)
    {
        found = 1;

        printf("\n--- Student Information Found ---\n");
        printf("ID:      %s\n",student[i].id);
        printf("Name:     %s\n",student[i].name);
        printf("Department: %s\n",student[i].department);
        printf("Semester:  %s\n",student[i].semester);
        printf("CGPA:     %.2lf\n",student[i].cgpa);

        int choice;

        printf("Are you sure you want to delete this student
information?\n");
        printf("1. Yes\n");
        printf("2. No\n");
        printf("Enter your choice: ");
        scanf("%d",&choice);

        if(choice == 1)
        {
            for(int j = i; j < *count - 1; j++)
```

```
        {
            student[j] = student[j + 1];
        }

        (*count)--;

        printf("Student information deleted
successfully!\n");
    }
    else if(choice == 2)
    {
        printf("Deletion canceled.\n");
    }
    else
    {
        printf("Invalid choice! Deletion canceled.\n");
    }

    break;
}

}

if(found == 0)
{
    printf("Student with ID %s not found.\n",deleteid);
}
}
```

```
void calculateGrade(STD student[],int count)
{
    if(count == 0)
    {
        printf("No student information available.\n");
        return;
    }

    char searchid[20];
    printf("\nEnter Student ID to see result: ");
    scanf("%s",searchid);

    int found = 0;

    for(int i = 0; i < count; i++)
    {
        if(strcmp(student[i].id,searchid) == 0)
        {
            found = 1;

            if(student[i].cgpa >= 3.75)
            {
                strcpy(student[i].grade,"A+");
            }
            else if(student[i].cgpa >= 3.50)
            {
```

```
        strcpy(student[i].grade,"A");
    }
    else if(student[i].cgpa >= 3.25)
    {
        strcpy(student[i].grade,"B+");
    }
    else if(student[i].cgpa >= 3.00)
    {
        strcpy(student[i].grade,"B");
    }
    else if(student[i].cgpa >= 2.75)
    {
        strcpy(student[i].grade,"C+");
    }
    else if(student[i].cgpa >= 2.50)
    {
        strcpy(student[i].grade,"C");
    }
    else if(student[i].cgpa >= 2.00)
    {
        strcpy(student[i].grade,"D");
    }
    else
    {
        strcpy(student[i].grade,"F");
    }
```

```

printf("\n=====
==\n");
        printf("          Academic Result\n");

printf("=====
\n");

        printf("ID: %s\n",student[i].id);
        printf("Name: %s\n",student[i].name);
        printf("Department: %s\n",student[i].department);
        printf("Semester: %s\n",student[i].semester);

        printf("-----\n");

        printf("CGPA: %.2lf\n",student[i].cgpa);
        printf("Grade: %s\n",student[i].grade);

        printf("-----\n");

        if(strcmp(student[i].grade,"F") == 0)
        {
            printf("Status: Failed\n");
        }
        else
        {
            printf("Status: Passed\n");

```

```

        }

printf("=====
\n");

        break;
    }
}

if(found == 0)
{
    printf("\nStudent with ID %s not found!\n",searchid);
}
}

void pauseScreen()
{
    printf("\nPress Enter to continue...");
    getchar();
    getchar();
}

int checkDuplicateID(STD student[],int count,char id[])
{
    for(int i = 0; i < count; i++)
    {

```

```
        if(strcmp(student[i].id,id) == 0)
        {
            return 1;
        }
    }

    return 0;
}

void saveFile(STD student[],int count)
{
    FILE *fp;

    fp = fopen("student.txt","w");

    if(fp == NULL)
    {
        printf("File cannot be opened!\n");
        return;
    }

    for(int i = 0; i < count; i++)
    {
        fprintf(fp,"%s\n",student[i].id);
        fprintf(fp,"%s\n",student[i].name);
        fprintf(fp,"%s\n",student[i].department);
        fprintf(fp,"%s\n",student[i].semester);
    }
}
```

```
        fprintf(fp,"%0.2lf\n",student[i].cgpa);
        fprintf(fp,"%s\n",student[i].grade);
    }

    fclose(fp);
}

void loadFile(STD student[],int *count)
{
    FILE *fp;

    fp = fopen("student.txt","r");

    if(fp == NULL)
    {
        return;
    }

    while(fscanf(fp,"%s\n",student[*count].id) == 1)
    {
        if(*count >= 100)
        {
            break;
        }

        fgets(student[*count].name,50,fp);
```

```
student[*count].name[strcspn(student[*count].name,"\n")]
= 0;

fgets(student[*count].department,30,fp);

student[*count].department[strcspn(student[*count].depart
ment,"\n")] = 0;

fscanf(fp,"%s\n",student[*count].semester);

fscanf(fp,"%lf\n",&student[*count].cgpa);

fgets(student[*count].grade,20,fp);

student[*count].grade[strcspn(student[*count].grade,"\n")]
= 0;

(*count)++;
}

fclose(fp);
}
```

7. Sample Input and Output

Sample Input:

Student ID : C261006
 Name : Mohammad Ferdous
 Department : CSE
 Semester : 1st
 CGPA : 3.80

Sample Output:

Student Added Successfully.

Student List

SL	ID	Name	Department	Semester	CGPA	Grade
1	C261006	Ferdous	CSE	1st	3.80	A+

Grade Calculation Reference Table:

CGPA Range	Grade
3.75 – 4.00	A+
3.50 – 3.74	A
3.25 – 3.49	B+
3.00 – 3.24	B
2.75 – 2.99	C+
2.50 – 2.74	C
2.00 – 2.49	D
Below 2.00	F

Screenshots:

Screenshot: Main Menu]

```
*****
Student Box
International Islamic University Chittagong
*****

=====
Main Menu
=====
1. Add Student Information
2. Display Student Information
3. Search Student Information
4. Update Student Information
5. Delete Student Information
6. Calculate Student Grade
7. Save & Exit
=====
Enter your choice: |
```

- Screenshot: Add Student

```

*****
                Student Box
International Islamic University Chittagong
*****

=====
                Main Menu
=====
1. Add Student Information
2. Display Student Information
3. Search Student Information
4. Update Student Information
5. Delete Student Information
6. Calculate Student Grade
7. Save & Exit
=====
Enter your choice: 1

--- Add Student Information ---
Enter ID: C261006
ID already exists! Please enter another ID.
Enter ID: C261015
Enter Name: Nasir
Enter Department: CSE
Enter Semester (1-12): 2nd
Enter CGPA (0.0 - 4.0): 3.55
Student information added successfully!

Press Enter to continue...|

```

-
-

[Screenshot: Display Student List]

```

*****
                Student Box
International Islamic University Chittagong
*****

=====
                Main Menu
=====
1. Add Student Information
2. Display Student Information
3. Search Student Information
4. Update Student Information
5. Delete Student Information
6. Calculate Student Grade
7. Save & Exit
=====
Enter your choice: 2

ID            Name            Department    Semester    CGPA
-----
C261006      Mohammad Ferdous      CSE          1st         2.54
C261008      Mohammad Numan        CSE          2nd         3.45
C261002      Raisul Rakib          CSE          1st         3.25
C261009      Abrar Siam            CSE          1st         3.12

Press Enter to continue...|

```

- [Screenshot: Search Student]

```
*****
                        Student Box
International Islamic University Chittagong
*****

=====
                        Main Menu
=====
1. Add Student Information
2. Display Student Information
3. Search Student Information
4. Update Student Information
5. Delete Student Information
6. Calculate Student Grade
7. Save & Exit
=====
Enter your choice: 3
Enter Student ID to search: C261006

--- Student Information Found ---
ID:          C261006
Name:        Mohammad Ferdous
Department:  CSE
Semester:    1st
CGPA:        2.54

Press Enter to continue...|
```

-
-

- [Screenshot: Update Student]

```

*****
                        Student Box
International Islamic University Chittagong
*****

=====
                        Main Menu
=====
1. Add Student Information
2. Display Student Information
3. Search Student Information
4. Update Student Information
5. Delete Student Information
6. Calculate Student Grade
7. Save & Exit
=====
Enter your choice: C261008
Enter Student ID to update:
--- Update Student Information ---
New Name: Mohammad Numan
New Department: CSE
New Semester (1-12): 1st
New CGPA (0.0 - 4): 3.45
Student information updated successfully!

Press Enter to continue...|

```

-
-
- [Screenshot: Delete Student]
-

```

*****
                Student Box
International Islamic University Chittagong
*****

=====
                Main Menu
=====
1. Add Student Information
2. Display Student Information
3. Search Student Information
4. Update Student Information
5. Delete Student Information
6. Calculate Student Grade
7. Save & Exit
=====
Enter your choice: 5
Enter Student ID to delete: C261009

--- Student Information Found ---
ID:          C261009
Name:        Abrar Siam
Department:  CSE
Semester:    1st
CGPA:        3.12
Are you sure you want to delete this student information?
1. Yes
2. No
Enter your choice: 1
Student information deleted successfully!

Press Enter to continue...|

```

-
-
-
-
-
- [Screenshot: Calculate Grade]

```
*****
                        Student Box
International Islamic University Chittagong
*****
```

```
=====
                        Main Menu
=====
1. Add Student Information
2. Display Student Information
3. Search Student Information
4. Update Student Information
5. Delete Student Information
6. Calculate Student Grade
7. Save & Exit
=====
```

Enter your choice: 6

Enter Student ID to see result: C261006

```
=====
                        Academic Result
=====
ID: C261006
Name: Mohammad Ferdous
Department: CSE
Semester: 1st
-----
```

CGPA: 2.54

Grade: C

```
-----
Status: Passed
=====
```

Press Enter to continue...|

-
-
- [Screenshot: Save and Exit]

```

*****
                Student Box
International Islamic University Chittagong
*****

=====
                Main Menu
=====
1. Add Student Information
2. Display Student Information
3. Search Student Information
4. Update Student Information
5. Delete Student Information
6. Calculate Student Grade
7. Save & Exit
=====
Enter your choice: 7

Are you sure you want to exit? (Y/N): y
Data saved successfully. Thank you

Process returned 0 (0x0)    execution time : 5.560 s
Press any key to continue.

```

-

8. Conclusion

The Student Box project successfully fulfills the requirements of a basic Student Management System. It provides a simple, efficient, and user-friendly way to manage student records.

Through this project, we learned how to use structures, arrays, functions, loops, file handling, and string manipulation in C programming. We also gained practical experience in designing a menu-driven application and solving real-world problems using programming.

The system stores student information permanently, making it more useful than a temporary program. This project has improved our programming knowledge and teamwork skills and will be helpful for future software development projects.

References

- C Programming Language by Brian W. Kernighan & Dennis M. Ritchie
- Code::Blocks IDE Documentation
- Class Lecture Notes
- Structured Programming Course Materials